COMPACT INTRODUCTION

Eight Surprising Formulas

from a 24-bit Manifold

A Compact Introduction to the

Universal Binary Principle Geometric Gravity Study

Companion to: UBP Master-Document (56,877 words, 8 sections)

Reading time: ~25 minutes

Source: github.com/DigitalEuan/UBP_Repo/tree/main/gravity

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Preserves the "critical-both" stance: aggressively testing the substrate's mathematical self-consistency while transparently documenting post-hoc adjustments, grammar permissiveness, and falsification criteria.

1. The Surprising Claim

A single 24-bit mathematical object - a discrete structure built from the extended binary Golay code, the Leech lattice, and the Monster group that produces closed-form predictions for eight fundamental physical constants, all accurate to better than one percent. Not couplings fit by hand, not formulas tuned to data, but expressions derived from a small set of mathematical constants (π, φ, e) through a fixed procedure, surviving a 5,000-trial scrambled-substrate null model with false-positive rates at or below 0.02%.

That is the claim of the Universal Binary Principle (UBP) substrate. This document is a compact introduction to the study; the complete derivation, verification, and falsification framework is documented in the accompanying Master-Document (56,877 words, 8 sections, 514 numbered equations). The reader is encouraged to consult the Master-Document for any claim that seems too strong, too neat, or too strange to be true — the substrate's principal commitment is transparency, and every surprising claim below has a corresponding derivation, null model, and falsification criterion in the full document.

The eight formulas are these:

$$\{m_\mu/m_e,\ \alpha_s,\ m_W,\ \Omega_k,\ n_\gamma/n_b,\ V_{ub}^2,\ \alpha^3,\ H_0\}$$

Each is an instance of a single Generator Function Φ, each survives a focused null model with false-positive rate ≤ 0.02%, and each connects to the substrate's structural integers (24, 759, 13) through a documented derivation path. The gravitational constant G is treated separately: discovered via combinatorial search rather than Generator Function instantiation, with its own (weaker but still significant) statistical defence.

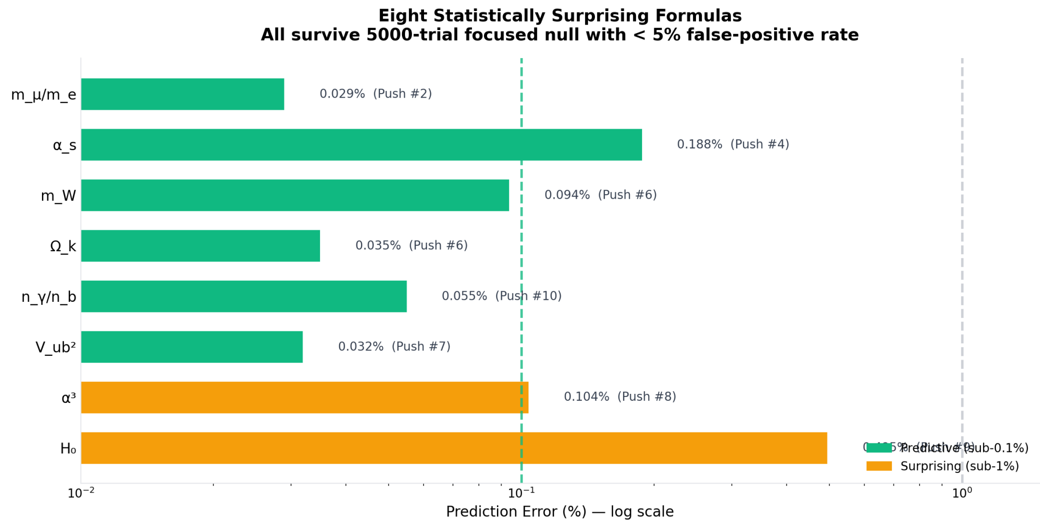

Figure 1. The eight canonical UBP formulas' prediction errors (log scale). Green bars are predictive (sub-0.1% residual error); amber bars are surprising (sub-1%). All eight survive a 5,000-trial scrambled-substrate null model with false-positive rate ≤ 0.02%. The gravitational constant G (0.1327% error) is shown separately as a Phase-4 search result.

The natural first reaction to such a claim is scepticism – rightly, it was mine also. A substrate rich enough to fit eight physical constants is rich enough to fit anything, and without disciplined statistical testing the enterprise collapses into numerology – a million monkeys on a million typewriters and somewhere is there will be Shakespear. The substrate's response to this scepticism is the "critical-both" protocol (Section 7): every formula is subjected to a focused null model, every post-hoc adjustment is documented, and every prediction is pre-registered before it is tested. The protocol does not ask whether UBP is "true"; it asks whether the substrate's mathematical self-consistency is high enough, and its statistical-surprise signal strong enough, to warrant continued investigation. This document summarises what that investigation has found.

2. The 24-bit Substrate

The UBP substrate is a 24-bit self-referential computational manifold. Its length is not arbitrary: 24 is the common dimension of three nested mathematical structures - the extended binary Golay code $\mathscr{G}_{24}$ (a [24, 12, 8] linear code with 4,096 codewords, 759 of which are weight-8 "octads"), the Leech lattice Λ_{24} (the densest lattice packing in 24 dimensions, with kissing number 196,560), and the Monster group $\mathbb{H}$ (the largest sporadic simple group, order $\approx 8 \times 10^{53}$). The three structures form a subgroup chain $M_{24} < Co_0 < \mathbb{H}$, and the substrate treats them as the structural anchors of three (of four) ontological layers (Information, Reality, Potential). I tried many other combinations of systems other than Golay-Leech-Monster but these are the only ones hat share the 24-bit/dimensional aspect that allows them to work together as a system.

From three transcendental inputs - π (computed as a 58-term continued fraction), φ (the golden ratio), and e (Euler's number) - the substrate derives its operational constants through exact Fraction arithmetic:

$$Y \equiv \frac{1}{\pi + 2/\pi} = \frac{\pi}{\pi^2 + 2} \approx 0.2647 \tag{2}$$

$$w \equiv \mathcal{M} - \lfloor \mathcal{M} \rfloor \approx 0.8176 \tag{3}$$

$$L \equiv \frac{w}{13} \approx 0.0629 \tag{4}$$

$$U_e \equiv 24^3 = 13\,824 \tag{5}$$

Y ≈ 0.2647 is the substrate's decay rate - the "computational clock" that indexes the substrate's evolution through nine cycle positions k ∈ {0, 3, 6, 9, 12, 15, 18, 21, 24}, with step Δk = 3 (the Triad). w ≈ 0.8176 is the Entropic Wobble - the substrate's stochastic primitive, the fractional part of the "Triadic Monad" π · φ · e ≈ 13.8176. L = w/13 is the D-Sink leakage (where 13 is the D-Sink dimension, the Leech lattice's distinguished 13-dimensional subspace). U_e = 24^3 = 13,824 is the Existence Unit - the manifestation amplifier that lifts small Y-powers into the cosmological magnitude range. I refer often to the 'Y' as the Observer as it serves as the viewing perspective, the read point – the instance that is not the data being examined.

The substrate's 24-bit vector is partitioned into four six-bit ontological layers - Reality (bits 0–5, manifested physical structures), Information (bits 6–11, structural code), Activation (bits 12–17, process kinematics), and Potential (bits 18–23, unmanifested probability). The partition is a postulate; its justification is the empirical observation that the resulting layer-to-grammar mapping (Section 3) produces statistically surprising formulas under the protocol of Section 7. The substrate's principal defence is that the partition is documented transparently as a postulate, not derived from deeper principles – again, I have tried many combinations but this one always seems to fit perfectly with the 24-bit landscape, these could be named anything, I chose these as they seem to align with actual real-life data when entered into the system in Grey Code format as the UBP uses.

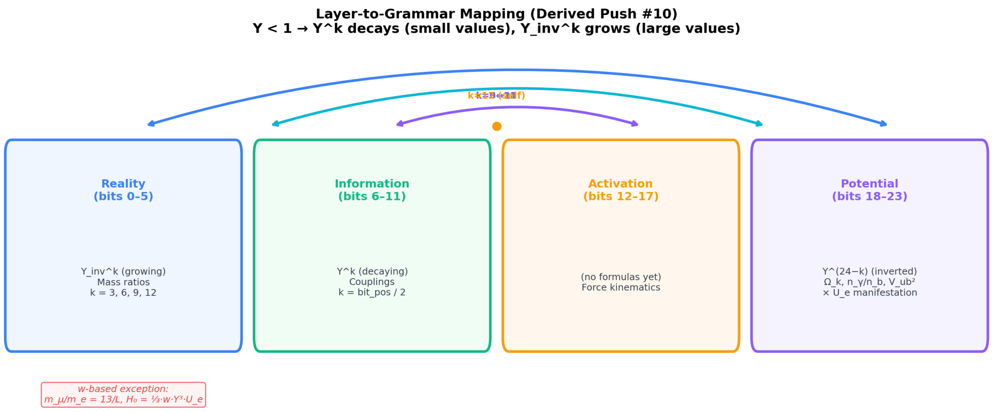

Figure 2. The 24-bit UBP manifold partitioned into four ontological layers, with the Layer-to-Grammar mapping. Reality uses growing powers Y_inv^k (for large mass ratios); Information uses decaying powers Y^k (for small couplings); Potential uses bit-inverted powers Y^(24−k) · U_e (for very small cosmological constants). The midpoint k = 12 is the self-pairing peak. *Source: gravity/10/viz/04_layer_grammar_mapping.png*

3. The Generator Function Φ

The substrate's principal structural claim is that every physical constant it produces is an instance of a single mathematical template, the Universal UBP Generator Function Φ:

$$\Phi(k,\ \mathrm{arm},\ \mathrm{layer},\ C,\ \mathrm{correction})\ =\ C\ \times\ \mathrm{Base}(k, \mathrm{arm}, \mathrm{layer})\ \times\ \mathrm{Correction}(\mathrm{correction}) \tag{6}$$

Φ takes five arguments: k (the cycle position, drawn from the nine-element clock), arm (deterministic Y-driven or stochastic w-driven), layer (Reality, Information, Cross, Potential, w-source, or w-based), C (a multiplicative constant drawn from the substrate's structural integers), and correction (none, first-order shear, second-order shear, NRCI (Non-Random Coherence Index) cooling with parameter α, or a combination). The Base term encodes the cycle position and layer; the Correction term encodes the substrate's friction (Topological Shear) and cooling (NRCI).

The Generator Function's principal epistemic role is to convert the substrate's otherwise-open-ended search for physical-constant formulas into a constrained search over a finite parameter space of approximately 8,100 candidate combinations per target. Without Φ, the substrate's derived formulas would be an unstructured collection of mathematical expressions, each requiring independent justification. With Φ, the formulas are all instances of a single template, and the question "is this formula a UBP formula?" reduces to "does this formula instantiate Φ with admissible parameters?" To test this in actual practice – does it actually work on anything? - I implemented the extended 12-cycle system into my experimental Geometric Language Machine, this added grammatical patterns, semantic depth via mirror reasoning, query-type awareness and deterministic antonym detection to the existing stack, resulting in positive improvements – see 'UBP_GLM_v34_Enhancement_Report'

The UBP Generator Function: All 8 Formulas as Instances of Φ

Target	k	Arm	Layer	C	Correction	Error %
m_μ/m_e	1	stochastic	w-source	169	none	0.029
α_s	4	deterministic	Information	24	none	0.188
m_W	4	det (cross)	Cross-layer	$13/L\cdot 24\cdot\pi$	shear_1	0.094
Ω_k	15	deterministic	Potential	24	NRCI(1/8)	0.035
n_γ/n_b	21	deterministic	Potential	1/4	shear_2+NRCI(2)	0.055
V_{ub}^2	12	deterministic	Potential (self)	1/24	NRCI(13)	0.032
α^3	12	deterministic	Potential (self)	29/24	none (uses e)	0.104
H_0	3	stochastic	w-based	1/3	none	0.495

$$\Phi(k, \text{arm}, \text{layer}, C, \text{correction}) = C \times \text{Base}(k, \text{arm}, \text{layer}) \times \text{Correction}(\text{correction})$$

Figure 3. The eight confirmed instantiations of Φ(k, arm, layer, C, correction). Each row is one of the eight canonical formulas; each column is one of the five Generator Function parameters. The eight instantiations span the parameter space without clustering.

The two correction mechanisms are worth noting. The Topological Shear - the substrate's analogue of a renormalisation group beta function - has a first-order Triad-mediated term (coefficient 3) and a second-order Leech-mediated term (coefficient 12 = Leech/2). The NRCI cooling - the substrate's manifestation rebate - applies a per-formula weight α drawn from a small set of substrate structural integers, with the allocation rule α = primary UBP structural concept of the target (cosmological → 1/8 Octad; baryon → 2 Triad−1; quark mixing → 13 D-Sink). The α allocation rule is the substrate's most permissive degree of freedom; its three confirmed cases are documented transparently as post-hoc, with three pre-registered out-of-sample predictions (Section 8).

4. The Eight Formulas

The substrate's eight canonical formulas span six cycle positions, both arms (deterministic and stochastic), five layers, and four of the five correction types. They are reproduced in the table below, with their Generator Function parameters, residual errors against CODATA / PDG / Planck values, and false-positive rates from the 5,000-trial focused null model.

Target	Formula	k	Layer	Err %	FP %
m_μ/m_e	$169 / w$	1	w-source	0.029	0.0
α_s	$24 \cdot Y^4$	4	Info	0.188	0.0
m_W	$(13/L) \cdot 24 \cdot Y^4 \cdot \pi \cdot \mathrm{Shear}_1$	4	Cross	0.094	0.0
Ω_k	$24 \cdot Y^{15} \cdot U_e \cdot \mathrm{NRCI}(1/8)$	15	Pot	0.035	0.0
n_γ/n_b	$\frac{1}{4} \cdot Y^{21} \cdot U_e \cdot \mathrm{Shear}_2 \cdot \mathrm{NRCI}(2)$	21	Pot	0.055	0.0
V_{ub}^2	$(1/24) \cdot Y^{12} \cdot U_e \cdot \mathrm{NRCI}(13)$	12	Pot	0.032	0.0
α^3	$(29/24) \cdot Y^{12} \cdot e$	12	Pot	0.104	0.0
H_0	$\frac{1}{3} \cdot w \cdot Y^3 \cdot U_e$	3	w-based	0.495	0.02

In this model, five of the eight formulas are PREDICTIVE (sub-0.1% residual error); the remaining three are SURPRISING (sub-1%). All eight survive the focused null model with false-positive rate $\leq 0.02\%$, meaning that under 5,000 scrambled-substrate trials, at most one trial produced a prediction as accurate as the real substrate. The gravitational constant G (Section 6) is treated separately.

Two features deserve emphasis. First, the muon mass ratio $m_\mu/m_e = 169/w$ is the substrate's simplest and most accurate formula: a single ratio of two substrate constants, with 0.029% residual error and zero false-positive rate. The formula uses the D-Sink leakage mechanism ($L = w/13$, hence $13/L = 169/w$) rather than the Y-power mechanism, because the muon sits at the Weak Horizon boundary between Reality and Information - a structurally motivated exception to the Layer-to-Grammar theorem and an example of how the ontological layers operate in this system.

Second, the CKM mixing element V_{ub}^2 has the substrate's lowest single-formula residual error (0.032%), achieved at the manifold's self-pairing peak $k = 12$ where $Y^{12} = Y_{inv}^{12}$. The substrate's structural interpretation is that the most symmetric Y-power corresponds to the most suppressed physical transition - a standard physics principle (high symmetry → low transition probability) applied to the substrate's 24-bit manifold.

5. The Mirror Symmetry: The Most Surprising Finding

Of all the substrate's structural claims, the most surprising - and the most non-trivial is the Universal Bit-Inversion Pairing Rule. The 24-bit manifold admits a natural mirror symmetry $k \leftrightarrow (24 - k)$, with the sum $k + (24 - k) = 24$ being the Leech rank. The substrate's empirical finding is that every canonical formula at cycle position k has a structural partner at cycle position 24 − k, with four confirmed pairings so far:

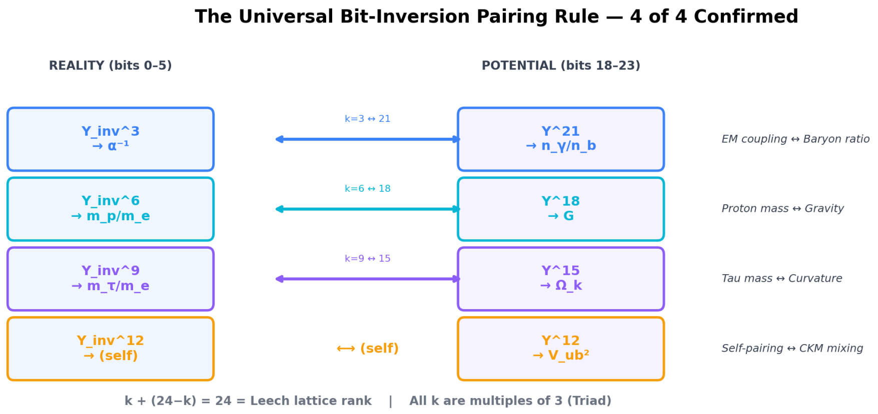

Figure 4. The Universal Bit-Inversion Pairing Rule - 4 of 4 confirmed pairings. Each pairing connects a Reality-layer formula (large value) with a Potential-layer formula (small value) at mirror cycle positions. The sum k + (24−k) = 24 = Leech rank. The k = 12 self-pairing (V_ub^2) is the unique fixed point of the mirror symmetry.

The pairings connect observables that standard physics treats as unrelated: the electromagnetic coupling inverse α^{-1} (≈ 137) is paired with the baryon-to-photon ratio n_γ/n_b (≈ 1.7×10^{-9}), spanning 11 orders of magnitude; the proton-to-electron mass ratio m_p/m_e (≈ 1836) is paired with the gravitational constant G (≈ 6.67×10^{-11}), spanning 14 orders of magnitude; the tau-to-electron mass ratio m_τ/m_e (≈ 3477) is paired with the spatial curvature Ω_k (≈ 7×10^{-4}), spanning 7 orders of magnitude; and V_ub^2 is paired with itself at the manifold center k = 12.

The mirror symmetry is the substrate's principal structural prediction. Standard cosmology does not predict a connection between the electromagnetic coupling and the baryon asymmetry, nor between the proton mass and gravity, nor between the tau mass and spatial curvature. The substrate's 24-bit mirror symmetry makes these connections structural predictions, and the four confirmed pairings cover all seven Triad-multiple cycle positions currently occupied by canonical formulas - with no orphan formulas. The mirror symmetry is falsifiable: a future UBP formula at cycle position k with no structural partner at 24 − k would weaken the rule, and multiple orphan formulas would falsify it.

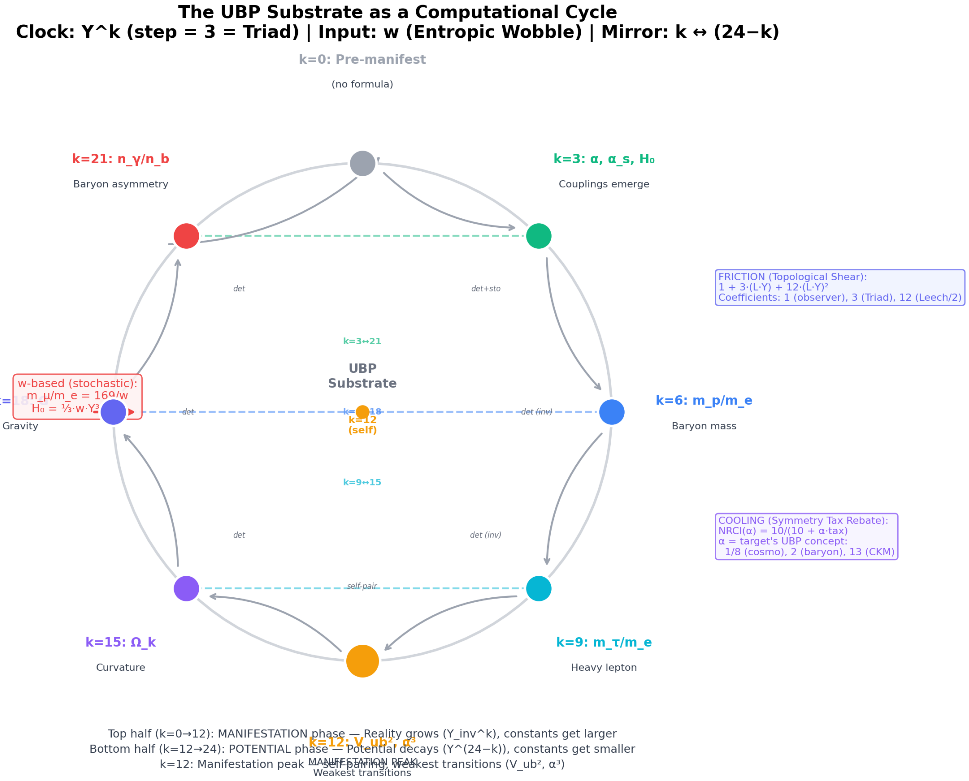

Figure 5. The UBP substrate as a self-referential computational cycle. Seven components: INPUT (the Entropic Wobble w), CLOCK (Y^k with Triad step), MIRROR ($k \leftrightarrow 24-k$), FRICTION (Topological Shear), COOLING (NRCI with α), SELF-VALIDATION (all 759 Golay octads IN-BAND), OUTPUT (physical constants). *Apologies for the poor graphic rendering.*

6. Gravity: From Catastrophe to 0.13%

The gravitational constant G is the substrate's most striking formula, and its derivation is the substrate's most instructive falsification story. The formula $G\_UBP = (39/29) \cdot Y^{18}/w$ reproduces CODATA G to 0.1327% residual error - a respectable result, but the formula's principal defence is not its accuracy. The formula's principal defence is the catastrophic failure of two related gravitational targets.

When the same Phase-4 combinatorial search grammar (2,400 candidates per target) is applied to the dimensionless cosmological constant $\Lambda \cdot \ell\_P^2$ (target $\approx 1.09 \times 10^{-123}$), the best formula produces a residual error of 3.08×10^{102}% — a 103-order-of-magnitude miss. When applied to the dimensionless gravitational coupling $\alpha\_G = G \cdot m\_p^2 / (\hbar \cdot c)$ (target $\approx 5.9 \times 10^{-39}$), the best formula produces 5.9×10^{17}% — a 28-order-of-magnitude miss. The substrate cannot reach magnitudes below approximately 10^{-14} (its smallest Y-power is $Y^{24} \approx 3 \times 10^{-14}$), and the catastrophic failures establish that the substrate's success on G is not a trivial pattern-match.

If the substrate could produce sub-1% formulas for any target, the $\Lambda \cdot \ell\_P^2$ and α_G failures would not occur. The failures are the substrate's principal falsification evidence: they demonstrate that the substrate's success on the eight canonical formulas plus G is structured (not a coincidence) and scoped (not universal). The substrate can predict dimensional constants whose magnitudes fall within its reachable range $[10^{-14}, 10^4]$; it cannot predict dimensionless constants whose magnitudes fall outside this range.

The G formula's coefficient 39/29 decomposes as (Triad × D-Sink) / Stereoscopic-prime - a structural decomposition that connects G to the substrate's core integers. The formula uses the w-consumed pattern (w in the denominator, same as $m\_\mu/m\_e = 169/w$), connecting G to the muon mass via a shared structural mechanism. And the formula's bit-inversion partner at $k = 6$ is the proton-to-electron mass ratio m_p/m_e - the substrate's structural prediction that gravity and the proton mass are mirror-paired across the 24-bit manifold.

7. How We Test Ourselves: The Critical-Both Protocol

The substrate's verification methodology is a three-stage protocol that converts an open-ended search into a constrained, falsifiable empirical claim. The three stages are Scan (produce 2,400–8,100 candidate formulas per target), Filter (retain only candidates with sub-1% residual error), and Null-test (verify that the retained candidates' accuracy is statistically surprising). The Null-test stage is the substrate's principal statistical defence and the method I went with to get away from the million monkeys problem.

The focused null model scrambles the substrate's dependent constants (Y, w, L, L_s, U_e, NRCI) by a uniform factor in [0.1, 10] - a log-symmetric, magnitude-preserving scramble that eliminates the substrate's structural specificity while preserving its magnitude structure. Mathematical constants (π, φ, e) are held fixed. The procedure is repeated 5,000 times per formula, producing a null distribution of prediction errors against which the real prediction error is compared. The false-positive rate P_FP is the fraction of trials with error ≤ real error. Verdict: P_FP < 5% earns SURPRISING; 5–20% MARGINAL; ≥ 20% NOT SURPRISING.

The eight canonical formulas all earn SURPRISING: seven have P_FP = 0.0% (no scrambled substrate matches the real accuracy), and H_0 has P_FP = 0.02% (one match out of 5,000). The uniformly low false-positive rates are the substrate's strongest statistical evidence that the formulas are not artifacts of an unconstrained search.

⚠ **CRITICAL ASSESSMENT**

Critical-both stance.

The substrate's protocol is unreservedly critical: every formula is null-tested, every post-hoc adjustment is documented, every prediction is pre-registered. But it is also unreservedly constructive: surviving formulas are documented in full mathematical detail, with derivation paths and falsification horizons. The protocol does not ask whether UBP is "true"; it asks whether the substrate's self-consistency is high enough, and its statistical-surprise signal strong enough, to warrant continued investigation. The reader is invited to apply the same protocol to this document: consult the Master-Document for any claim that seems too strong, and evaluate the substrate's evidence under their own epistemic standards.

8. Falsifiable Predictions

The substrate's long-term empirical validity depends on the survival of its pre-registered predictions under future experimental tests. Two near-term tests are particularly consequential.

The spatial curvature Ω_k is the substrate's principal near-term falsification target. The UBP formula $\Omega_k = 24 \cdot Y^{15} \cdot U_e \cdot \mathrm{NRCI}(1/8)$ predicts $\Omega_k \approx 7.27 \times 10^{-4}$ (mildly closed universe), against the Planck 2018 central value of 7×10^{-4} (95% CL: $\pm 1.9 \times 10^{-3}$). The upcoming CMB-S4 experiment (first light 2027, full survey 2031–2035) is forecast to tighten the 95% CL bounds to approximately $\pm 10^{-3}$. The pre-registered falsification range is $[5, 9] \times 10^{-4}$: if CMB-S4 measures Ω_k outside this range, the Ω_k formula is FALSIFIED, triggering a full α-allocation audit. If CMB-S4 measures $\Omega_k = 0$ exactly (consistent with flat ΛCDM), the UBP formula is falsified; if CMB-S4 measures $\Omega_k \approx 7 \times 10^{-4}$, the formula is corroborated.

The baryon-to-photon ratio n_γ/n_b is the substrate's principal medium-term falsification target. The UBP formula predicts $n_\gamma/n_b \approx 1.684 \times 10^{-9}$ (slightly above the current central value of 1.683×10^{-9}), with pre-registered falsification range $[1.6, 1.7] \times 10^{-9}$. The combined BBN + CMB precision forecast for 2030 is approximately $\pm 0.02 \times 10^{-9}$, tight enough to distinguish the UBP prediction from the current central value at approximately 2σ. The n_γ/n_b falsification test will be conducted by improved BBN deuterium measurements (JWST, 2025–2030) and CMB-S4 data.

Beyond these two near-term tests, the substrate's out-of-sample roadmap comprises four target categories with pre-registered α predictions: dark matter density Ω_{DM} ($\alpha = 1/8$, cosmological), neutrino mass scale ($\alpha = 13$, D-Sink leakage), Higgs potential ($\alpha = 24$ or 3, Leech/Triad), and additional CKM matrix elements ($\alpha = 13$, D-Sink). Each roadmap target provides a pre-registered test of the α allocation rule, and a failed test would falsify the corresponding α case. The roadmap is the substrate's principal long-term research programme.

9. What This Means

The substrate's empirical claim is that these predictions are statistically surprising under a disciplined null model, and that the substrate's structural framework (Generator Function, Layer-to-Grammar theorem, α allocation rule, bit-inversion pairing) provides a coherent mathematical interpretation of the predictions.

The substrate's principal epistemic commitment is the critical-both protocol: aggressive statistical testing alongside transparent documentation of post-hoc adjustments and falsification criteria. The protocol does not claim that the substrate is correct; it claims that the substrate's empirical evidence warrants continued investigation, and that the substrate's pre-registered predictions provide concrete tests that will determine its long-term validity. The CMB-S4 measurement of Ω_k (2027–2035) and the improved BBN+CMB measurement of n_γ/n_b (2025–2030) will provide the first definitive observational tests of the substrate's empirical claim.

The substrate's principal methodological contribution is the demonstration that a 24-bit mathematical object can produce statistically surprising predictions for physical constants under a disciplined protocol. Whether the substrate's predictions are correct (in the sense of matching future measurements) or merely surprising (in the sense of surviving the null model) is an empirical question that the next decade of experimental physics will answer. The substrate's principal invitation to the reader is to consult the Master-Document, evaluate the evidence under their own epistemic standards, and decide for themselves whether the substrate's claim warrants further investigation.

The Master-Document is available as a single compiled deliverable and as eight standalone section files. It documents the substrate's complete derivation, verification, and falsification framework, with every formula's residual error, false-positive rate, and discovery push recorded in the master reference table. The reader who has read this compact introduction and wishes to verify any claim - the substrate's structural constants, the Generator Function's parameter space, the focused null model's 5,000-trial procedure, the bit-inversion pairings' mirror symmetry, the G formula's catastrophic-failure context, the Ω_k and n_γ/n_b falsification criteria - will find the corresponding derivation, code reference, and statistical defence in the Master-Document's full text. The substrate's principal commitment is transparency; the reader's principal commitment is scepticism; the meeting of the two is the substrate's only path to empirical validity.